SSUET/QR/112

Laboratory Manual

Introduction to Operating Systems

(CS-222L)

4rd Semester Spring-2024

Student Name: ________________________________

Student Roll# : __________________ **Section :** _____

BS in Computer Science

Department of Computer Science & Information Technology
Sir Syed University of Engineering and Technology
University Road Karachi - 75300
http://www.ssuet.edu.pk

Course Learning Outcome(s):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Understanding the basic concepts of Operating Systems	PLO_1 (Engineering Knowledge)	C2 (Understand)
2	Examine and design small-scale combinational operating System's commands	PLO_1 (Engineering Knowledge)	P2,C5 (Set)
3	Perform Practical lab individual or in group.	PLO_9 (Individual & Teamwork)	A2,C3 (Respond to Phenomena)

Laboratory Experiments

Lab No.	Topics	Signature
1	Introduction to Linux operating system, Installation of Linux operating system in a Virtual Machine and explore its features.	
2	Managing Directories, Hardware Information and Kernel distribution information, Redirection and pipes, Filters commands in Linux.	
3	File /directory Permission and Process Management with System Calls commands.	
4	Introduction of vi text editor to create and edit files	
5	Implementation of Process and thread (Life cycle of process): (i) Process creation and Termination; (ii) Thread creation and Termination	
6	Implementation of CPU Scheduling Algorithm (i)FCFS (First Come First Serve) (ii) SJF (Shortest Job First)	
7	Implementation of CPU Scheduling Algorithm (i)Round Robin (ii)Priority based	
8	Implementation of Deadlock Avoidance Banker's Algorithm and study about deadlock recovery	
9	Producer-Consumer Problem using Semaphores	
10	Implementation of Memory Management using (i)First Fit (ii)Worst Fit & (iii) Best Fit	
11	Execute the Linux Shell scripts	
12	Execute the Shell script by using Control Structures and Loops	
13	(a)Execute the Shell script by using Switch Case (b)Execute the Shell script by using Functions	
14	Introduction to Docker, its working with benefits and Docker hub account creation.	

Laboratory Rubrics

Criteria	Exceeds Expectations (>=90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)
Software Handling (2)	Able to use software with its standard and advanced features without assistance	Able to use software with its standard and advanced features with minimal assistance	Able to use software with its standard features with assistance	Unable to use the software
Programming/ Simulation (5)	Able to program/ simulate the lab tasks with simplification	Able to program/ simulate the lab tasks without errors	Able to program/ simulate lab tasks with errors	Unable to program/simulate
Results (2)	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results
Lab Report (1)	Laboratory manual has no grammatical and/ or spelling errors. All sections of the report are very well written and technically accurate.	Laboratory manual has very few grammatical/ spelling errors. All sections of the report are technically accurate.	-Laboratory manual has multiple grammatical/ spelling errors. Few sections of the report contain technical error s.	Laboratory manual has several grammatical/ spelling errors and sentence construction is poor. -All sections of the report contain multiple technical errors.

Laboratory Report Score

Lab#	Criteria				Score
1.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
2.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
3.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
4.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
5.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
6.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
7.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
8.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
9.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
10.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
11.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
12.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
13.	Software Handling ( )/2	Programming/ Simulations ( )/5	Results ( )/2	Lab Report ( )/1	
14.	Software Handling ( )/4	Programming/ Simulations ( )/5	Results ( )/4	Lab Report ( )/2	
Obtained Score out of (140)					
Obtained Score out of (X) = Obtained Score out of (140)/140 * X					

Semester Project Rubric

Criteria	Exceeds Expectations (>=90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)	Score
Project Demonstration (5)	Able to demonstrate the project with achievement of required objectives having clear understanding of project limitations and future enhancements. Hardware and/or Software modules are fully functional, if applicable.	Able to demonstrate the project with achievement of required objectives but understanding of project limitations and future enhancements is insufficient. Hardware and/or Software modules are functional, if applicable.	Able to demonstrate the project with achievement of a*t least 50% required objectives and insufficient understanding of project limitations and future enhancements. Hardware and/or Software modules are partially functional, if applicable.	Able to demonstrate the project with achievement of less than 50% required objectives and lacks in understanding of project limitations and future enhancements. Hardware and/or Software modules are not functional, if applicable.	
Project results (3)	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results	
Report Writing (3)	Project report has no grammatical and/ or spelling errors. All sections of the report are very well-written and technically accurate.	Project report has very few grammatical/ spelling errors. All sections of the report are technically accurate.	Project report has multiple grammatical/ spelling errors. Few sections of the report contain technical errors.	Project report has several grammatical/ spelling errors and sentence construction is poor.	
Viva (4)	Able to answer the questions easily and correctly across the project.	Able to answer the questions related to the project	Able to answer the questions but with mistakes	Unable to answer the questions	
				Obtained Score out of (15)	
				Obtained Score out of (X) = Obtained Score out of (15)/15*X	

Laboratory Examination Rubric

Criteria	Exceeds Expectations (>=90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)	Score
Performance (15)	Able to present full knowledge of both problem and solution.	Able to present adequate knowledge of both problem and solution	Able to present sufficient knowledge of both problem and solution	No or very less knowledge of both problem and solution	
Viva (5)	Able to answer the questions easily and correctly across the project.	Able to answer the questions related to the project	Able to answer the questions but with mistakes	Unable to answer the questions	
				Obtained Score out of (20)	
				Obtained Score out of (X) = Obtained Score out of (20)/20*X	

Final Lab Assessment

Criteria	Score
Laboratory Report **(15)**	
Semester Project **(15)**	
Laboratory Examination **(20)**	
Total Score out of (50)	

Examiner: *Waqas Akber*

(Name and Signature)